

Credit Card Approval Prediction

Project Description:

The Credit Card Approval Prediction System is a machine learning-based web application designed to assist financial institutions in making fast, accurate, and reliable credit card approval decisions. The platform analyzes an applicant's demographic, financial, and employment information to predict credit card eligibility using a trained Random Forest classifier. It also generates a detailed risk audit report that explains the prediction and stores prediction history for future reference.

Built using Flask, Python, Scikit-learn, Pandas, NumPy, HTML, CSS, and JavaScript, the application provides an intuitive web interface for applicants and bank officials. The system performs data preprocessing, feature engineering, machine learning-based prediction, and maintains prediction logs in JSON format while supporting Docker-based deployment for scalability.

Scenarios

Scenario 1:

A customer applying for a credit card enters personal, employment, and financial information through the web application. The system preprocesses the data, predicts whether the application is likely to be approved, and displays the approval status along with a detailed risk audit report.

Scenario 2:

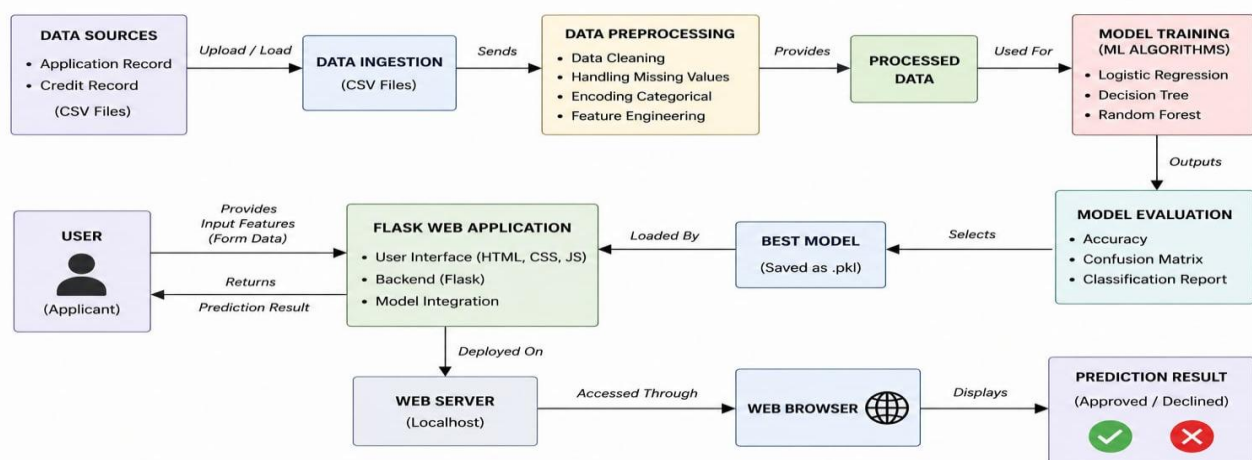
A bank credit officer uses the application to evaluate multiple credit card applications efficiently. The system analyzes each applicant's information using the trained Random Forest model and provides consistent approval predictions, helping reduce manual effort and decision-making time.

Scenario 3:

A financial institution integrates the Credit Card Approval Prediction System into its approval workflow to automate applicant evaluation. The system predicts credit card approval in real time, stores prediction history for future analysis, and helps reduce financial risk by improving the accuracy and consistency of approval decisions.

Technical Architecture

TECHNICAL ARCHITECTURE: CREDIT CARD APPROVAL PREDICTION



Description:

The Credit Card Approval Prediction System follows a modular three-tier architecture.

- The Frontend Layer consists of a Flask-based web interface developed using HTML, CSS, JavaScript, and Jinja2 templates.
 - The Backend Layer is implemented using Flask and handles request processing, routing, data validation, and machine learning prediction.
 - The Service Layer contains reusable modules for data preprocessing, feature engineering, Random Forest prediction, risk audit report generation, and prediction history management.
- The architecture enables scalability, maintainability, modular development, and efficient machine learning-based decision-making.
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Pre-requisites

- Python Programming Knowledge
 - Flask Framework Knowledge
 - HTML, CSS, and JavaScript Basics
 - Machine Learning Concepts
 - Scikit-learn Library
 - Data Preprocessing Techniques
 - Git and GitHub Basics
 - Virtual Environment Management
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Project Workflow**Milestone 1: Environment Setup and Architecture Design****Activity 1.1: Define Project Requirements**

- Identify user requirements.
- Define credit card approval use cases.
- Finalize project scope.

Activity 1.2: Design System Architecture

- Create architectural diagrams.
- Define interactions between frontend, backend, machine learning model, and data storage.

Activity 1.3: Set Up Development Environment

- Install Python.
- Create virtual environment.
- Install dependencies.

Activity 1.4: Create Project Structure

Create folders for:

- static/
 - templates/
 - model/
 - utils/
 - logs/
 - tests/
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Milestone 2: Backend Development**Activity 2.1: Implement Flask Backend**

Create application routes:

- /

- /predict
- /result
- /history

Responsibilities:

- Handle user requests.
 - Process applicant information.
 - Return prediction results.
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Activity 2.2: Develop Data Preprocessing Module

Implement preprocessing and feature engineering.

Functions:

`preprocess_data()`

Responsibilities:

- Validate applicant information.
 - Convert and normalize input values.
 - Prepare model input features.
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Activity 2.3: Develop Prediction Module

Functions:

`predict_approval()`

Responsibilities:

- Load the trained Random Forest model.
 - Predict credit card approval status.
 - Generate prediction confidence.
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Activity 2.4: Develop Risk Audit Module

Functions:

`generate_audit_report()`

Responsibilities:

- Generate approval summary.
 - Display risk-related information.
 - Explain prediction results.
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Milestone 3: Frontend Development

Activity 3.1: Develop Flask User Interface

Create user-friendly pages for:

- Applicant Information Form
 - Prediction Result
 - Risk Audit Report
 - Prediction Summary
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Activity 3.2: Add User Interaction Features

Implement:

- Form Validation
- Responsive Design
- Prediction Summary Display
- Error Handling

Milestone 4: Data Storage and Logging

Activity 4.1: Prediction History

Store prediction results in:

prediction_summary.json

Responsibilities:

- Save prediction history.
- Retrieve previous predictions.

Activity 4.2: Model Management

Store trained model as:

random_forest_model.joblib

Responsibilities:

- Load trained model.
- Perform prediction efficiently.

Milestone 5: Testing and Validation

Activity 5.1: Unit Testing

Create test files:

- test_prediction.py
- test_preprocessing.py
- test_routes.py

Use:

pytest -v

Validate:

- Prediction functionality.
- Data preprocessing.
- Flask routes.
- Model integration.

Activity 5.2: Application Testing

Run Flask Application:

<http://127.0.0.1:5000>

Test:

- Applicant form.
- Prediction functionality.
- Risk audit report.

Milestone 6: Deployment and Documentation

Activity 6.1: Local Deployment

Run Flask Application:

python app.py

Run using Docker:

docker-compose up

Activity 6.2: Version Control

Upload project to GitHub.

Repository includes:

- Source code
- Documentation
- Images
- Model files
- README

Exploring the Application Pages

Home Page

Description:

The homepage allows users to:

- Enter applicant information.
- Provide financial details.
- Predict credit card approval.

Prediction Result Page

Description:

Displays the machine learning prediction.

Example:

Approved

Approval Probability

Prediction Summary

Risk Audit Report

Description:

Displays a detailed explanation of the prediction along with applicant risk analysis and important evaluation factors.

Prediction History

Description:

Displays previously generated prediction records stored in the application.

Conclusion

The Credit Card Approval Prediction System successfully demonstrates how Machine Learning can improve the credit card approval process by providing fast, consistent, and accurate eligibility predictions. The system assists financial institutions by automating applicant evaluation, reducing manual effort, generating detailed risk audit reports, and maintaining prediction history for future analysis.

The modular architecture built using Flask and Scikit-learn ensures maintainability, scalability, and efficient deployment. Future enhancements include cloud deployment, user authentication, database integration, real-time model retraining, advanced explainable AI techniques, and support for multiple machine learning models.

This project highlights the practical application of Machine Learning in improving financial decision-making, reducing operational costs, and enhancing customer experience.